Table 19: The implementation algorithm of the proposed PPU_8 in the signed array multiplier.

Step	Operation	Equivalent Logic
1	$S_1 = 0$	$FALSE(S_1)$
2	$S_2 = 0$	$FALSE(S_2)$
3	$(C_{in} \rightarrow 0) \equiv (C_{in} \rightarrow S_1) = S_1{}'$	$S_1{}' = \overline{C_{in}}$
4	$(\beta \rightarrow S_1{}') = S_1{}''$	$S_1{}'' = \bar{\beta} + \overline{C_{in}}$
5	$(\beta \rightarrow 0) \equiv (\beta \rightarrow S_2) = S_2{}'$	$S_2{}' = \bar{\beta}$
6	$(S_2{}' \rightarrow C_{in}) = C_{in}{}'$	$C_{in}{}' = \beta + C_{in} = \boldsymbol{C_{out}}$
7	$S_2 = 0$	$FALSE(S_2)$
8	$(S_1{}'' \rightarrow 0) \equiv (S_1{}'' \rightarrow S_2) = S_2{}'$	$S_2{}' = \beta \cdot C_{in}$
9	$(C_{in}{}' \rightarrow S_2{}') = S_2{}''$	$S_2{}'' = \bar{\beta} \cdot \overline{C_{in}} + \beta \cdot C_{in} = \overline{\beta \oplus C_{in}} = \boldsymbol{Sum}$

4 Simulation results, evaluation, and comparison

This section of the article has been presented in two subsections. The first subsection presents the simulation results of the proposed structures at the circuit level. It also includes a comprehensive comparison of the area, energy consumption, and computational delay of the proposed multipliers with previous works, along with an evaluation of the results. In the second part, the functionality of the proposed multipliers is assessed in the applications of Gaussian blur and edge detection. The results of hardware implementation and computational complexity of these applications based on the proposed multipliers are compared with SOA.

4.1 Circuit simulation and its results

Various SPICE models have been presented to simulate memristive circuits. Voltage ThrEshold Adaptive Memristor (VTEAM) model is one of the nonlinear memristive models with high computational accuracy and low computational complexity [1, 31, 32]. This research applied the VTEAM model introduced at *TU Wien* to simulate the proposed circuits [4–6, 20–22, 28, 30]. The specifications of the applied memristor model and the IMPLY logic's design parameters are detailed in Table 20. In the circuit-level simulation, the pulse width of each computational step is considered equal to 30 μs, as reported in [4–6, 20–22, 28, 30].

Table 20: Setup values of IMPLY logic and VTEAM model [4, 5, 22, 28, 30].

Parameter	v_{set}	v_{reset}	v_{cond}	t_{pulse}	R_G	v_{off}	v_{on}	α_{off}	α_{on}
Value	1 V	1 V	900 mV	30 μs	40 KΩ	0.7 V	-10 mV	3	3
Parameter	R_{off}	R_{on}	k_{off}	k_{on}	w_{off}	w_{on}	w_C	a_{off}	a_{on}
Value	1 MΩ	10 KΩ	1 $\frac{cm}{s}$	-0.5 $\frac{nm}{s}$	0 nm	3 nm	107 pm	3 nm	0 nm

The proposed PPUs introduced in the last section and other PPUs of unsigned and signed array multipliers are simulated in the SPICE simulator using the mentioned VTEAM model. All the possible input states of the proposed circuits are simulated to ensure their correct functionality. For example, an unsigned array multiplier's PPU_1 has four inputs, so 16 input states must be assessed. All the outputs of PPUs were calculated correctly in all cases based on the simulation results. Only the simulation waveforms of one input state of the proposed unsigned array multiplier's $PPU_{1\text{-}3}$ are presented in Figures 4–6 due to the multiplicity of PPUs and their input states.

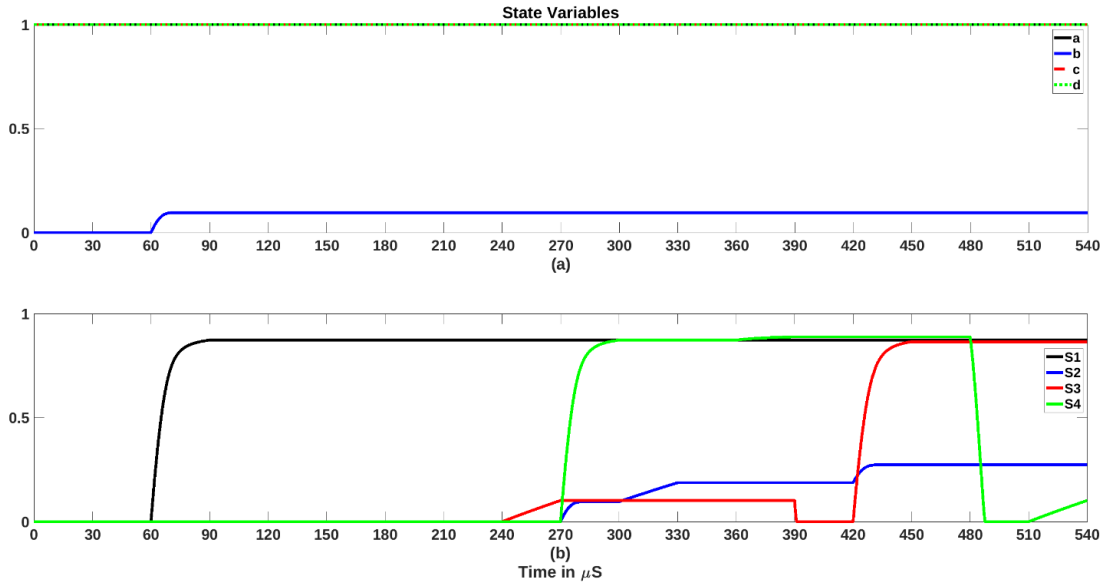

Figure 4: Simulation waveforms of the proposed unsigned array multiplier's PPU_1 for input state "1011".

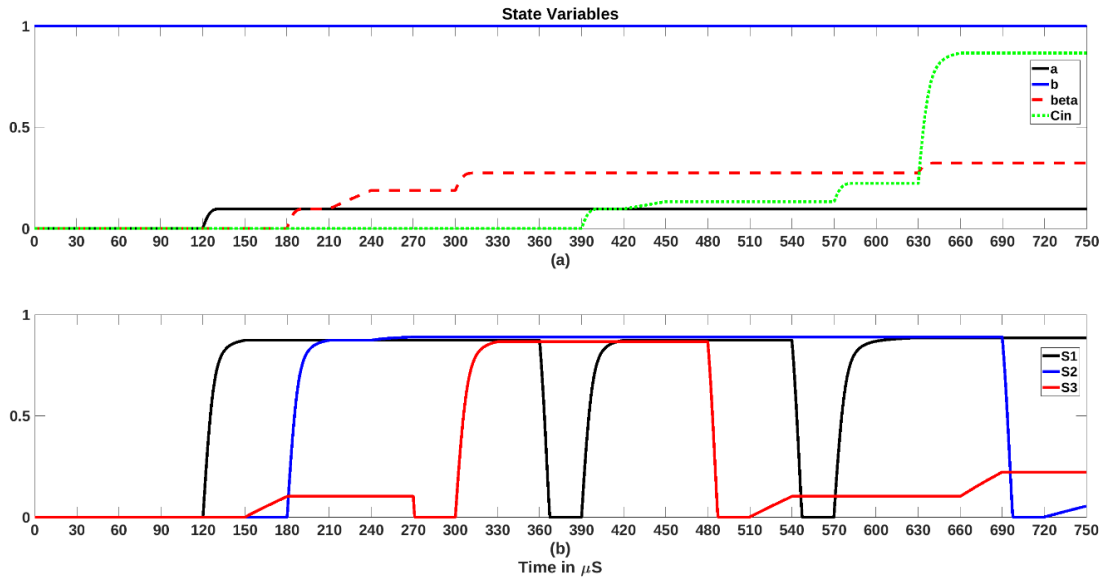

Figure 5: Simulation waveforms of the proposed unsigned array multiplier's PPU_2 for input state "0001".

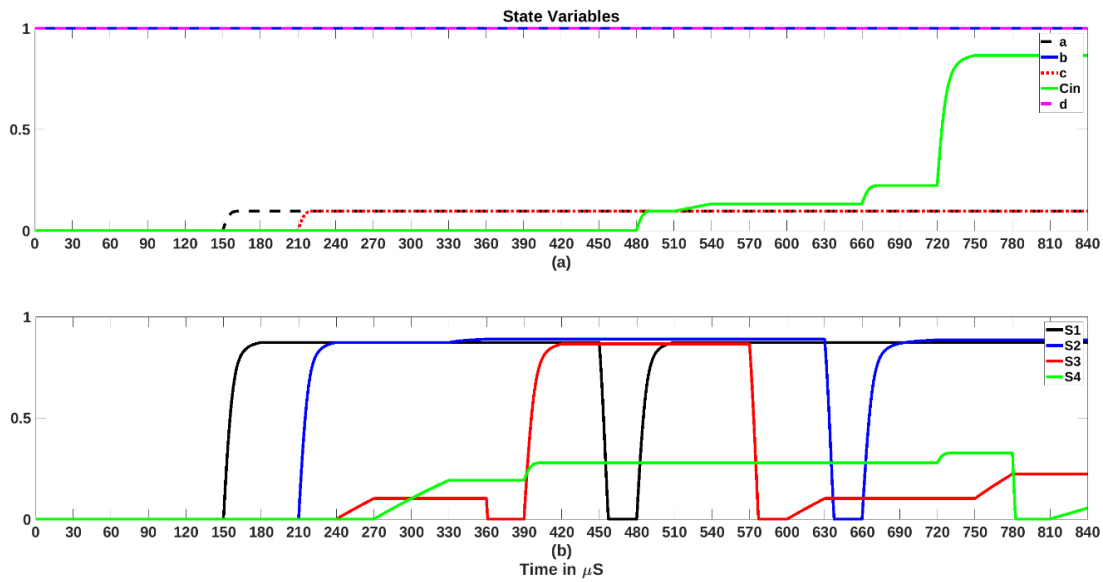

Figure 6: Simulation waveforms of the proposed unsigned array multiplier's PPU_3 for input state "00101".

The output waveform of the proposed signed array multiplier's PPUs (PPU_2 and PPU_7) for one of their input states are illustrated in Figures 7 and 8.

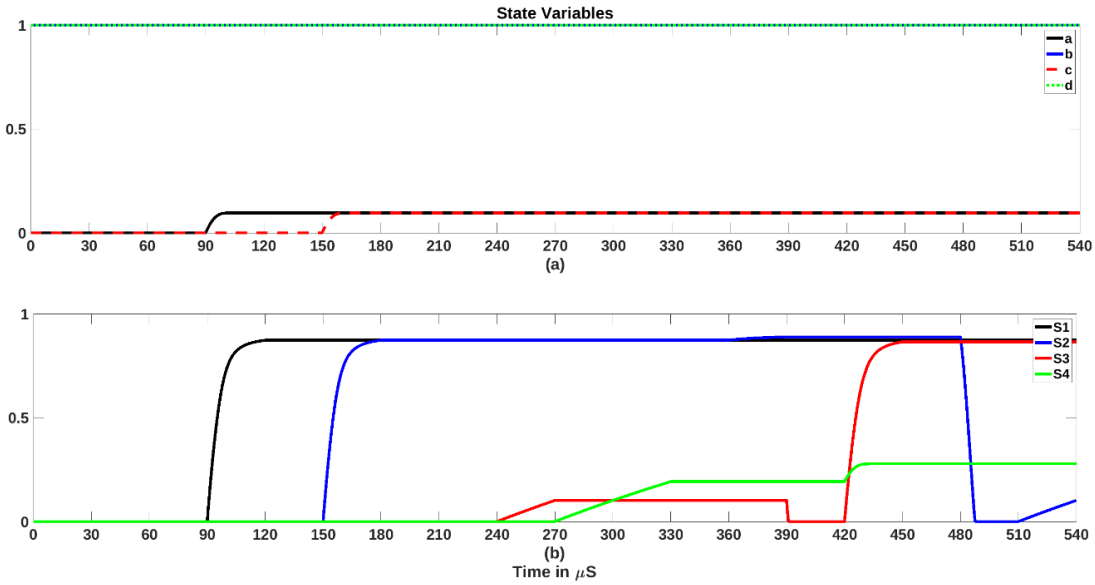

Figure 7: Simulation waveforms of the proposed signed array multiplier's PPU_2 for input state "0101".

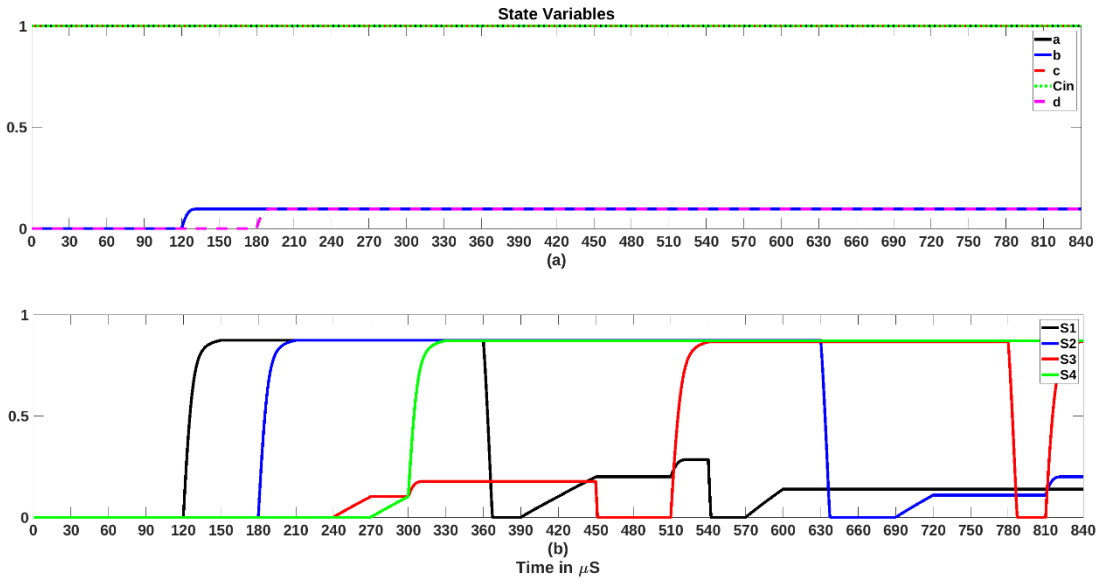

Figure 8: Simulation waveforms of the proposed signed array multiplier's PPU_7 for input state "11010".

Proposed $PPU_{1\text{-}3}$ of the unsigned array multiplier were implemented in 18, 25, and 28 computational steps, simulated in 0 µs–540 µs, 0 µs–750 µs, and 0 µs–850 µs, respectively. 450 µs–480 µs, 660 µs–690 µs, and 750 µs–780 µs are the intervals to compute the *Sum* output of the proposed $PPU_{1\text{-}3}$, respectively, in the unsigned array multiplier structure. The C_{out} output is computed and stored at the last computational step in all three proposed PPUs of this multiplier. Proposed PPU_2 and PPU_7 of the signed array multiplier are simulated in the time intervals of 0 µs–540 µs (18 computational steps) and 0 µs–840 µs (28 computational steps), respectively. The *Sum* output of the proposed PPU_2 and PPU_7 is simulated in the intervals of 450 µs–480 µs and 750 µs–780 µs, respectively, and each PPU's C_{out} output is calculated and stored in the last computational step.

The method introduced in [4, 5, 28, 30] was applied to calculate the energy consumption of each cell of the proposed unsigned and signed array multipliers. In Table 21, the estimated energy consumption of each component of the proposed unsigned and signed array multipliers is written along with the constituent units of other multipliers (such as the multiplexer in the radix-2 Booth multiplier and the 4:2 compressor proposed in [28]). The energy consumption of unsigned and signed array multipliers has been calculated by applying (26) and (27), respectively. $E_{UAmultiplier}$ and $E_{SAmultiplier}$ refer to the energy consumption of the unsigned and signed array multipliers in these equations. According to (26) and (27), the estimated energy consumption of n-bit unsigned and signed array multipliers equal $(2.156n^2 - 2.672n - 0.022)$ nJ and $(2.156n^2 - 2.703n - 0.067)$ nJ, respectively.